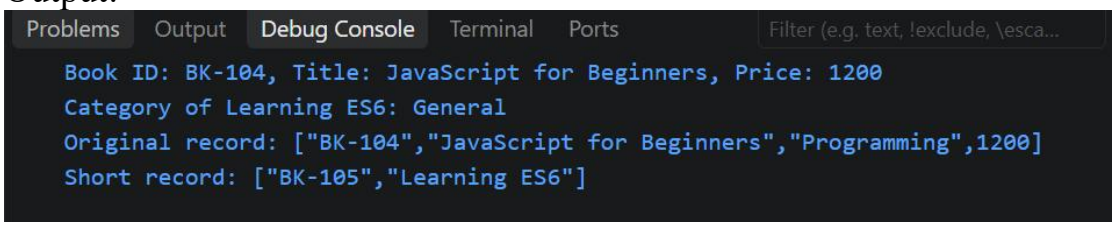
Question 1 - Library Book Record

Q1.JS

```
const bookRecord = ["BK-104", "JavaScript for Beginners", "Programming",
1200];
// Extract ID, title, skip category, and extract price
const [bookId, bookTitle, , bookPrice] = bookRecord;
// Shorter record: use General if category is missing
const shortRecord = ["BK-105", "Learning ES6"];
const [, , category = "General"] = shortRecord;
// Display book details using a template literal
console.log(`Book ID: ${bookId}, Title: ${bookTitle}, Price:
${bookPrice}`);
console.log(`Category of ${shortRecord[1]}: ${category}`);
// Original arrays remain unchanged
console.log("Original record:", bookRecord);
console.log("Short record:", shortRecord);
```

Output:

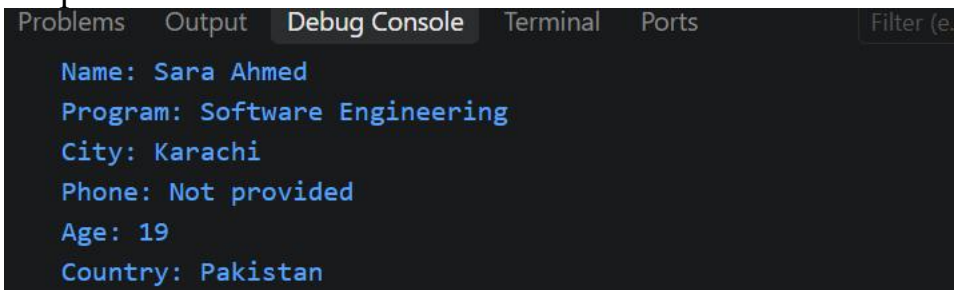


Question 2 - Student Profile from an API

Q2.JS

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Library Book Record</title>
</head>
<body>
    <h1>Library Book Details</h1>
    <script src="Question2.js"></script>
</body>
</html>
```

Output:



Question 3 - Manage Class Lists and Student Profiles

Q3.JS

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Library Book Record</title>
</head>
<body>
    <h1>Library Book Details</h1>
    <script src="Question3.js"></script>
</body>
</html>
```

Output:



Question 4 - Build a Reusable Student Toolkit with Modules

index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Student Toolkit</title>
</head>
<body>
    <h2>Student Toolkit</h2>
    <pre id="output"></pre>
    <script type="module" src="./main.js"></script>
</body>
</html>
```

studentToolkit.js:

```
function getFullName(firstName, lastName) {
    return `${firstName} ${lastName}`;
}
function calculateTotal(marks1, marks2, marks3, marks4) {
    return marks1 + marks2 + marks3 + marks4;
}
// Named export
export { getFullName }
// Default export
export default calculateTotal;
```

Main.js:

```
import calculateTotal, { getFullName } from "./studentTools.js";
const fullName = getFullName("Sara", "Ahmed");
const totalMarks = calculateTotal(85, 90, 78, 92);
const output = `Full Name: ${fullName}
Total Marks: ${totalMarks}`;
console.log(output);
document.getElementById("output").textContent = output;
```

Output:



Student Toolkit

Full Name: Sara Ahmed
Total Marks: 345

Question 5 - Mini Project: Course Registration Data

index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Course Registration</title>
</head>
<body>
    <h2>Course Registration</h2>
    <pre id="output"></pre>

    <script type="module" src="./main.js"></script>
</body>
</html>
```

Courses.js:

```
const courses = [
    {
        courseName: "Web Development",
        instructor: "Mr. Ali",
        duration: "3 months",
        fee: 15000
    },
    {
        courseName: "Python Programming",
        instructor: "Ms. Sara",
```

```
        duration: "2 months",
        fee: 12000
    },
    {
        courseName: "Graphic Design",
        instructor: "Mr. Ahmed",
        duration: "2 months",
        fee: 10000
    }
];
export default courses;
```

courseUtils.js:

```
function formatCourse(courseName, instructor, fee) {
    return `${courseName} is taught by ${instructor}. Course fee: Rs.
${fee}.`;
}
export default formatCourse;
```

Main.js:

```
import courses from "./courses.js";
import formatCourse from "./courseUtils.js";
// Extract course name and fee
const { courseName, fee } = courses[0];
console.log("Selected Course:", courseName);
console.log("Course Fee:", fee);
// Add a new course without changing the original list
const newCourse = {
    courseName: "Data Science",
    instructor: "Ms. Ayesha",
    duration: "4 months",
    fee: 20000
};
const updatedCourses = [...courses, newCourse];
// Update course fee without changing the original object
const updatedCourse = {
    ...courses[0],
    fee: 18000
};
// Prepare output
let output = `Selected Course: ${courseName}
Course Fee: Rs. ${fee}
Original Courses:
`;
courses.forEach(course => {
    output += `${course.courseName} - Rs. ${course.fee}\n`;
});
output += "\nUpdated Course List:\n";
updatedCourses.forEach(course => {
    output += `${course.courseName} - Rs. ${course.fee}\n`;
});
output += `\nUpdated Web Development Fee: Rs. ${updatedCourse.fee}\n`;
output += `Original Web Development Fee: Rs. ${courses[0].fee}\n`;
output += "\nCourse Summaries:\n";
```

```
updatedCourses.forEach(course => {
    output += formatCourse(
        course.courseName,
        course.instructor,
        course.fee
    ) + "\n";
});
console.log(output);
document.getElementById("output").textContent = output;
```

Output:

